

Semantics of Comparative Constructions

Class 3: Differentials, quantifiers in *than*-clauses

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OVA Summer School · University of Debrecen

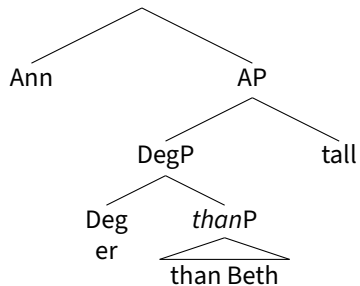
Part 1

Recap & today's plans

Recap of the syntax of comparatives, I

1) The *than*-clause is generated as the complement of *er*:

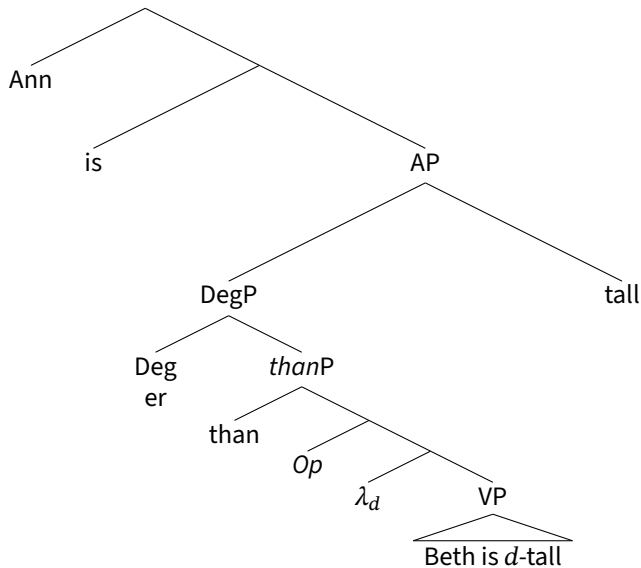
(1)



Recap of the syntax of comparatives, II

2) there's ellipsis & movement in the *than*-clause:

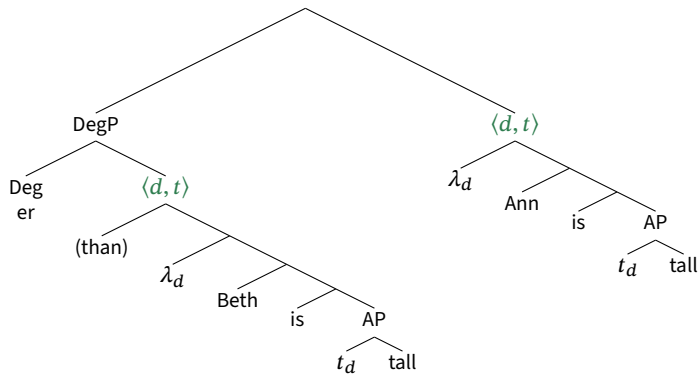
(2)



Recap of the syntax of comparatives, III

3) DegP headed by *er* QRs in the matrix clause:

(3)



Recap of the semantics of comparatives

- We assume er denotes a relation between degree intervals:

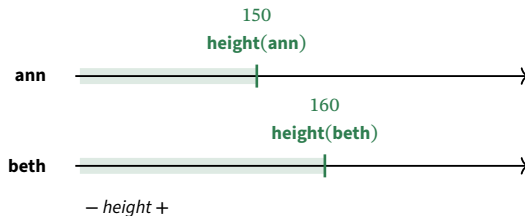
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- Example:



- (5) a. Ann is taller than Beth.
b. $\{d \mid \mathbf{tall}(\mathbf{beth}) \geq d\} \subset \{d \mid \mathbf{tall}(\mathbf{ann}) \geq d\}$

The commensurability condition

- We proposed that er has the following definedness condition:

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(7) * This movie is longer than Ann is tall.

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- Remaining issue:

(10) * Ann is taller than Beth isn't short.

Today's plan

- 1) side remarks on phrasal comparatives
- 2) a new (gradable) entry for *er*
- 3) we begin talking about the interpretation of quantifiers in *than*-clauses

Part 2

Phrasal comparatives

Phrasal comparatives in English

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- (13) a. Who is Ann taller than *t*?
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a. No man is stronger than himself.
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- (13) a. Who is Ann taller than *t*?
b. * Who is Ann taller than *t* is?
- (14) a. Ann is taller than no student.
b. * Ann is taller than no student is.

(Hankamer 1973, Hoeksema 1983, a.o.)

Phrasal & clausal comparatives in Greek I

- The debate concerning whether English in fact has two types of *than*Ps is far from settled.
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(e.g., Lechner 2001, Bhatt and Takahashi 2011)
- However, there are languages that have morphologically distinguish the two strategies.
- Consider, for example, Greek: (Merchant 2009)

- (15) a. I Maria pezi kiθara kalitera [**ap’oti** (pezi kiθara) o Giannis].
the Maria.NOM plays guitar better than.CLAUSAL plays guitar the Giannis.NOM
‘Maria plays the guitar better than Giannis plays the guitar.’
- b. I Maria pezi kiθara kalitera [**apo** ton Giannis].
the Maria.NOM plays guitar better than.PHRASAL the Giannis.ACC
‘Maria plays the guitar better than Giannis.’

(see Pancheva 2006 for Russian and other Slavic languages; see Hankamer 1973 for Hungarian)

Phrasal & clausal comparatives in Greek II

- In *ap'oti* comparatives, the case of the DP must match its main clause correlate; in *apo* comparatives, the complement must be in the accusative:

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- Similar contrasts to what we see in English:

- (17) a. * O Giannis den ine psiloteris [**ap'oti** kanenas].
the Giannis not is taller than.CLAUSAL n-person
Lit. 'Giannis isn't taller than anyone.'
- b. O Giannis den ine psiloteris [**apo** kanenan].
the Giannis not is taller than.PHRASAL n-person
'Giannis isn't taller than anyone.'

An analysis of phrasal comparatives

- We can analyze phrasal comparatives as involving a different version of *er*:

$$(18) \quad \llbracket \text{er}_{\text{phrasal}} \rrbracket = \lambda x_e. \lambda A_{\langle d, \langle e, t \rangle \rangle}. \lambda y_e. \{d \mid A(d)(x)\} \subset \{d \mid A(d)(y)\}$$

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- Illustration:

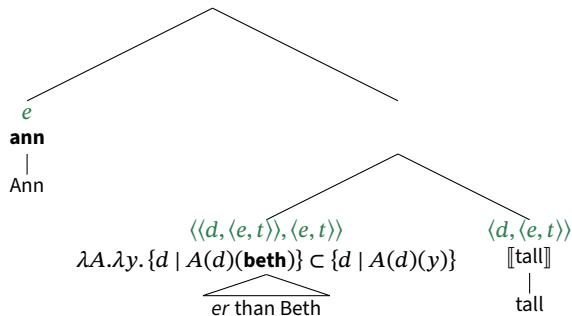
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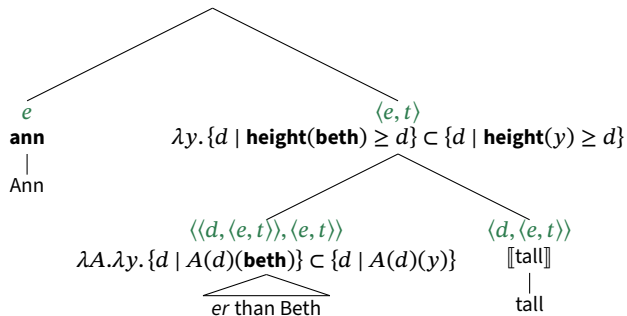
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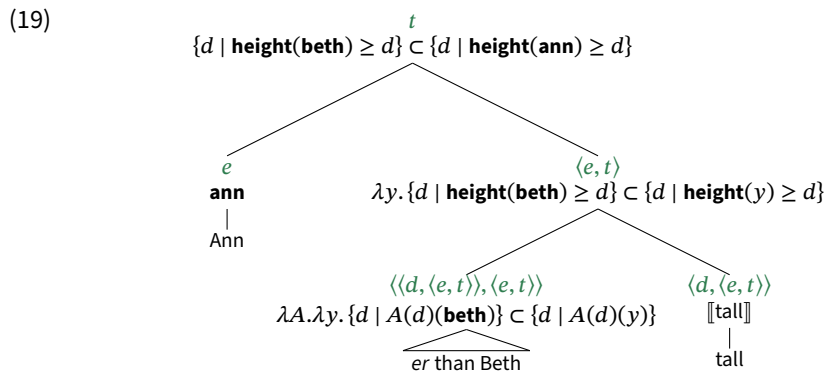


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Part 3

Measuring difference

Modifying comparatives

- Consider the following sentences:
(20) a. Ann is **5cm** taller than Beth.
b. Ann is **two Einsteins** smarter than Beth.
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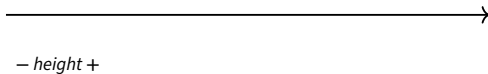
- (21) Ann is much taller than Beth than Cleo.

- Our entry for *er* doesn't allow degree modification, however. What do we need to do?

- (22) $\llbracket \text{er} \rrbracket = \lambda D. \lambda D'. D \subset D'$

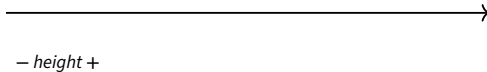
A new meaning for *er*

- We must add a degree argument to comparatives.
- But what would these degrees measure?



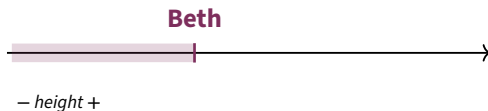
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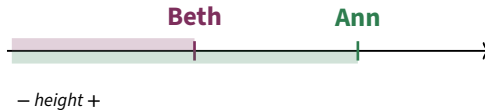
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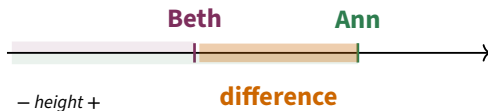
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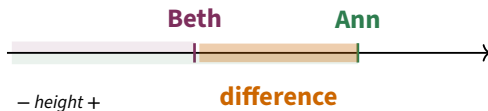
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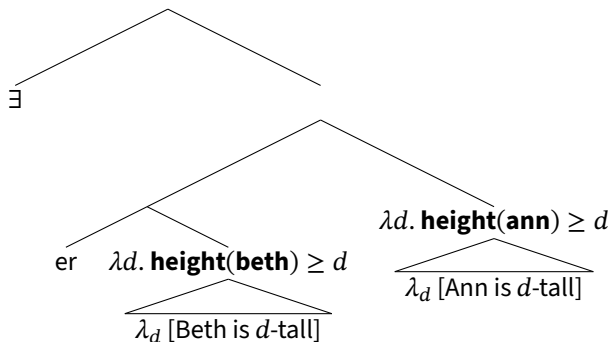
- Comparatives as set subtraction/difference:

$$\begin{aligned}(23) \quad \llbracket \text{er} \rrbracket &= \lambda D. \lambda D'. D - D' \\ &= \lambda D. \lambda D'. \lambda d. D(d) \wedge \neg D'(d)\end{aligned}$$

Sentences with no differentials

- When there's no modifier, the degree argument of the comparative is existentially closed:

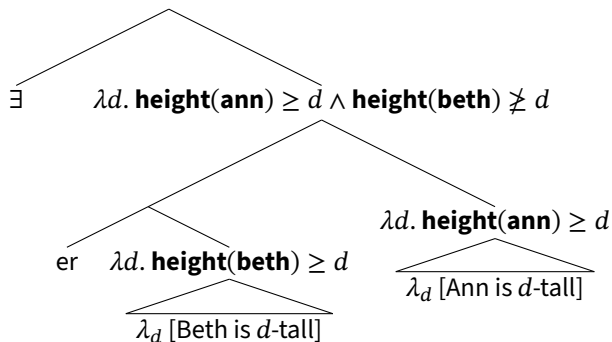
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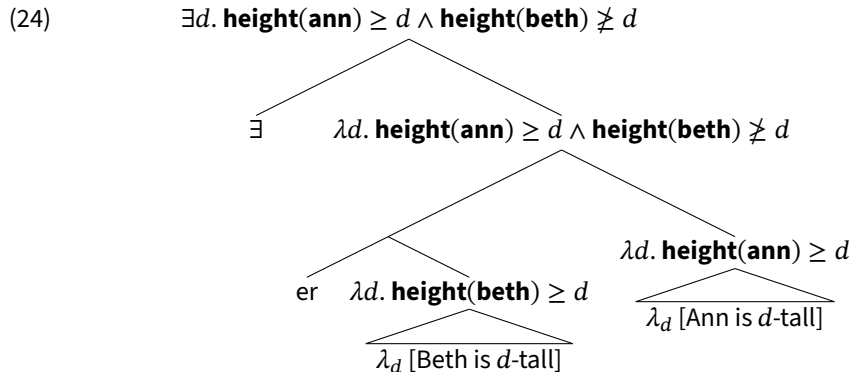
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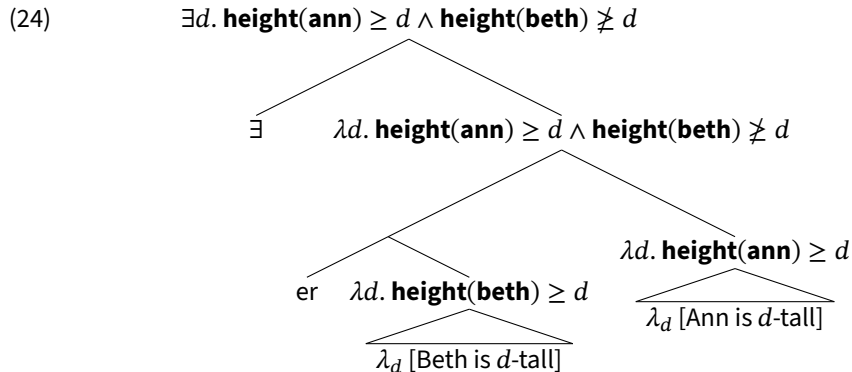
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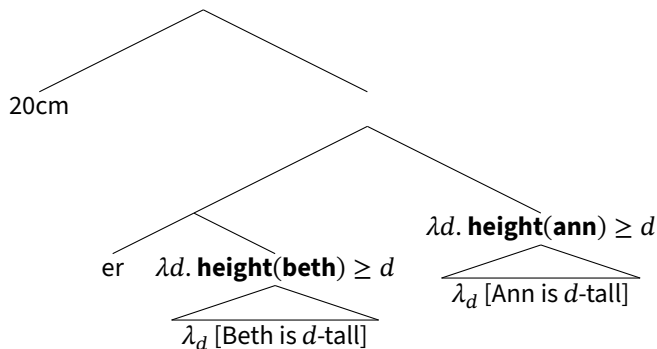


- The resulting analysis we get is often referred to as **A-not-A analysis**.
(Seuren 1973, Schwarzschild 2008)

Sentences with differentials

- When there's a differential, it will measure the difference between the two sets:

(25)

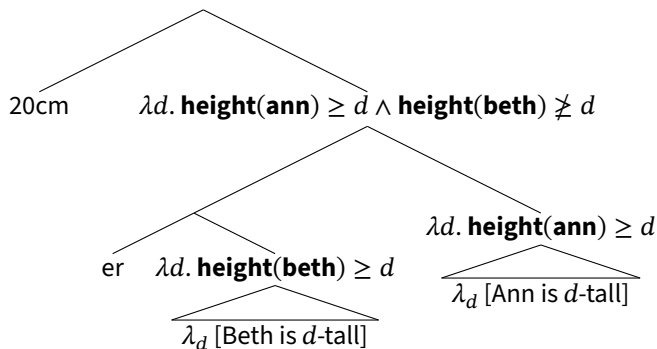


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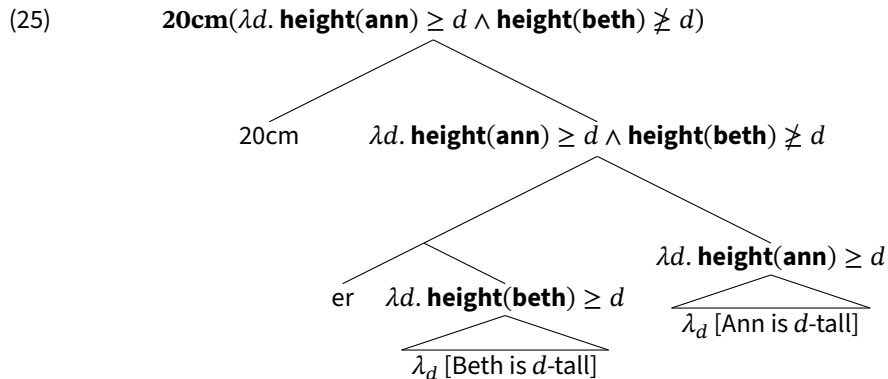
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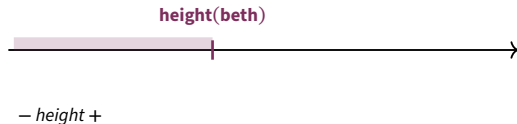


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A recipe for interpreting comparatives

1. Construct Beth's interval:

$$B = \{d \mid \mathbf{height}(\mathbf{beth}) \geq d\}$$



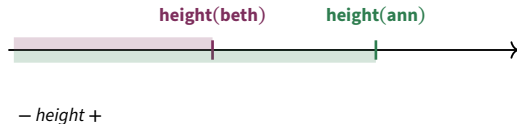
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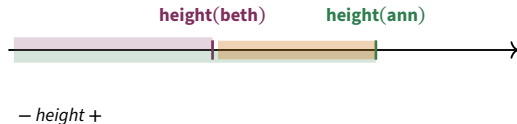
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2. Construct Ann's interval:

$$A = \{d \mid \text{height}(\text{ann}) \geq d\}$$

3. Subtract Beth's interval from Ann's:

$$A - B = \{d \mid \text{height}(\text{ann}) \geq d \wedge \text{height}(\text{beth}) \not\geq d\}$$



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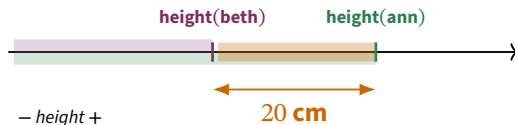
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3. Subtract Beth's interval from Ann's:

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4. Measure the resulting interval:

$$20\text{cm}(A \setminus B)$$



Part 4

Disjunction in *than*-clauses

The puzzle

- In our first class, we discussed the following puzzle:
- (26) Ann is taller than either Beth or Cleo are.
 $\rightsquigarrow$ *Ann is taller than Beth & Ann is taller than Cleo*

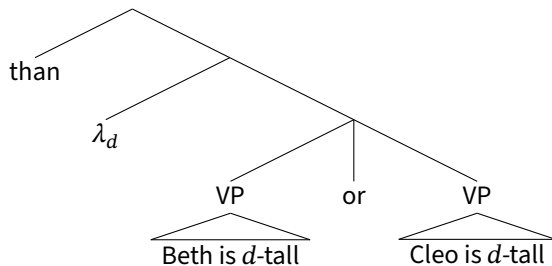
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(26) Ann is taller than either Beth or Cleo are.
 $\rightsquigarrow$ *Ann is taller than Beth & Ann is taller than Cleo*
- We are now in a position to account for this fact!

than-clauses with disjunction

- Consider the following *than*-clause:

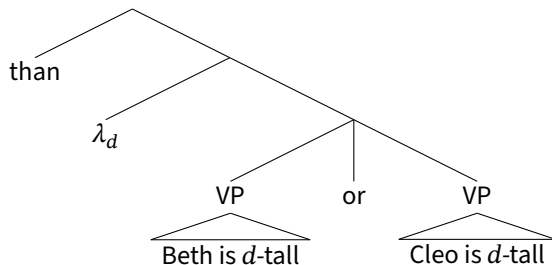
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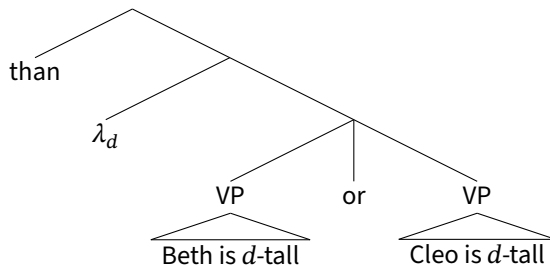
- This interval will denote the **union** of the Beth- and the Cleo-interval:

(28) $\llbracket \text{thanP} \rrbracket = \lambda d. (\mathbf{height}(\mathbf{beth}) \geq d) \vee (\mathbf{height}(\mathbf{cleo}) \geq d)$

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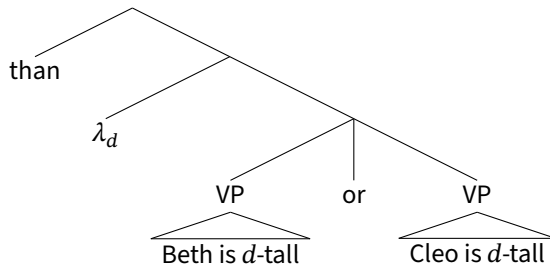
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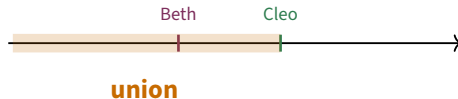
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Interpreting the full sentence



- Interpreting the full sentence:

- (29) a. Ann is taller than either Beth or Cleo are.
b. $\exists d. \text{height}(\text{ann}) \geq d \wedge \neg((\text{height}(\text{beth}) \geq d) \vee (\text{height}(\text{cleo}) \geq d))$

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Part 5

NPIs in *than*-clauses

NPIs in comparatives

- Negative polarity items (NPIs) are licensed in *than*-clauses:

- (30) a. I didn't read **any** books.
b. * I read **any** books.
- (31) a. Every girl that read **any** books passed the exam.
b. * Every girl that passed the exam read any books.
- (32) Ann is taller than any boy in her class.
 $\rightsquigarrow$ *Ann is taller than every boy in her class.*

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↗ *Ann is taller than every boy in her class.*

- NPIs are said to be licensed in **downward-entailing environments**.

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(31) a. Every girl that read **any** books passed the exam.

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(32) Ann is taller than any boy in her class.

$\rightsquigarrow$ *Ann is taller than every boy in her class.*

- NPIs are said to be licensed in **downward-entailing environments**.
- It should be straightforward to see that the A-not-A analysis predicts NPIs to be licensed in *than*-clauses (and not in matrix clauses):

(33) Ann is taller than Beth $\rightsquigarrow \exists d. \llbracket \text{tall} \rrbracket(d)(\mathbf{ann}) \wedge \neg \llbracket \text{tall} \rrbracket(d)(\mathbf{beth})$

NPIs in comparatives

- Negative polarity items (NPIs) are licensed in *than*-clauses:

(30) a. I didn't read **any** books.

b. * I read **any** books.

(31) a. Every girl that read **any** books passed the exam.

b. * Every girl that passed the exam read any books.

(32) Ann is taller than any boy in her class.

$\rightsquigarrow$ *Ann is taller than every boy in her class.*

- NPIs are said to be licensed in **downward-entailing environments**.
- It should be straightforward to see that the A-not-A analysis predicts NPIs to be licensed in *than*-clauses (and not in matrix clauses):

(33) Ann is taller than Beth $\rightsquigarrow \exists d. \llbracket \text{tall} \rrbracket(d)(\mathbf{ann}) \wedge \neg \llbracket \text{tall} \rrbracket(d)(\mathbf{beth})$

(though keep in mind that our previous entry for *-er* also captures these facts)

NCl in comparatives

- **Negative concord items** are sometimes licensed in *than*-clauses:

- (34) a. Eu não vi **ninguém**.
I not saw nobody
'I didn't see anybody.'
- b. * Eu vi **ninguém**.
I saw nobody
'I didn't see anybody.'
- c. * Todo mundo que viu ninguém gritou.
everyone who saw nobody screamed
- (35) Ele quer vencer mais do que **ninguém**.
he wants to win more of the than nobody.
'I wants to win more than anyone.'

Brazilian Portuguese

NClS in comparatives

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- Perhaps this suggests that maybe there's indeed a covert negation in comparatives...

NCIs in comparatives

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Brazilian Portuguese

- Perhaps this suggests that maybe there's indeed a covert negation in comparatives...
- But in Brazilian Portuguese, at least, these sentence are not that productive:

- (36) * Ele quer vencer mais do que **nenhum** aluno.
he wants to.win more of.the that no student.

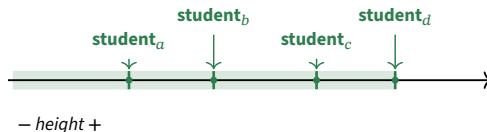
Interpreting than clauses with NPIs

- NPIs are often analyzed as simple indefinites:

$$(37) \quad \llbracket \text{any student} \rrbracket = \lambda P_{\langle e,t \rangle}. \exists x \in \mathbf{student}. P(x)$$

- Thus, *than*-clauses with NPIs will be interpreted just like those with disjunction:

$$(38) \quad \llbracket \lambda_d \text{ any student is } d \text{ tall} \rrbracket = \lambda d. \exists x \in \mathbf{student}. \mathbf{tall}(x) \geq d$$



Part 6

Modals in *than*-clauses

Allow in than-clauses

- (39) **Homework:** Write a paper between 2–4 pages. I wrote a 5-page paper.
My paper is longer than it is allowed to be.

Allow in than-clauses

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↪ *my paper is longer than 4 pages*

Allow in *than*-clauses

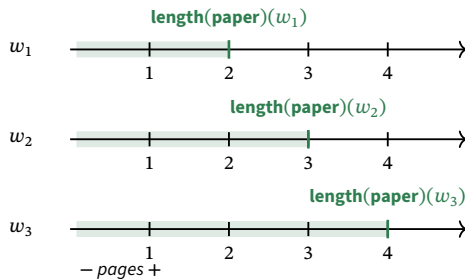
(39) **Homework:** Write a paper between 2–4 pages. I wrote a 5-page paper.

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$\rightsquigarrow$ *my paper is longer than 4 pages*

- Possibility modals like *allow* are existential quantifiers over possible worlds:

(40) $\llbracket \lambda_d \text{ the paper is allowed to be } d \text{ long} \rrbracket^@ = \lambda d. \exists w \in \mathbf{Rule}(@). \mathbf{long}(w)(\mathbf{the.paper}) \geq d$



Allow in *than*-clauses

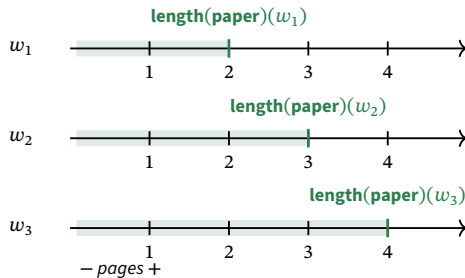
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$\Rightarrow$ The interval extends from 0 to 4 pages.

Require in than-clauses

(41) **Homework:** Write a paper between 2–4 pages. I wrote a 3-page paper.

My paper is longer than it needs to be.

Require in *than*-clauses

(41) **Homework:** Write a paper between 2–4 pages. I wrote a 3-page paper.

My paper is longer than it needs to be.

↪ *my paper is longer than 2 pages*

Require in *than*-clauses

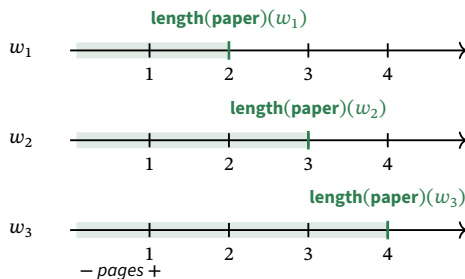
(41) **Homework:** Write a paper between 2–4 pages. I wrote a 3-page paper.

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$\rightsquigarrow$ *my paper is longer than 2 pages*

- Necessity modals like *need* are universal quantifiers over possible worlds:

(42) $\llbracket \lambda_d \text{ the paper needs to be } d \text{ long} \rrbracket^@ = \lambda d. \forall w \in \mathbf{Rule}(@). \mathbf{long}(w)(\mathbf{the.paper}) \geq d$



Require in *than*-clauses

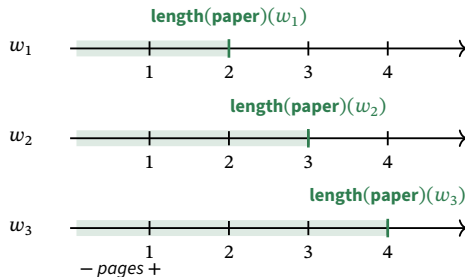
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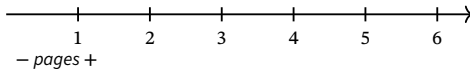
$\Rightarrow$ The interval extends from 0 to 2 pages.

Part 7

A puzzle concerning differentials

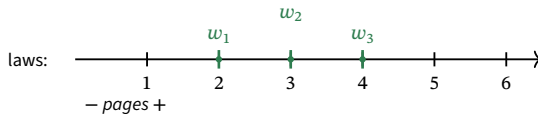
Differentials + *allow* in *than*-clauses

- (43) **Homework:** Write a paper between 2–4 pages. I wrote a 6-page paper.
My paper is 2 pages longer than it is allowed to be.



Differentials + *allow* in *than*-clauses

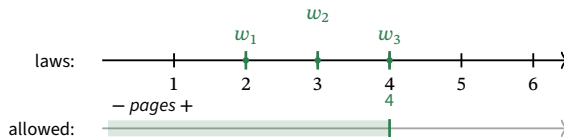
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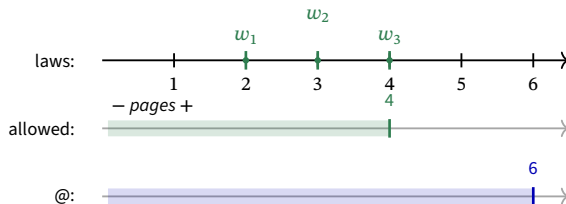
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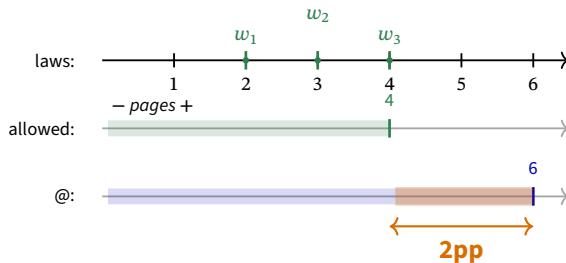
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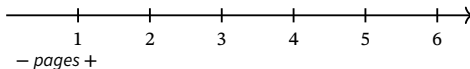
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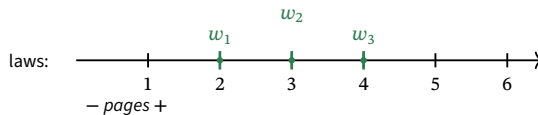
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Differentials + *need* in *than*-clauses

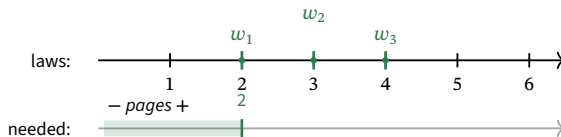
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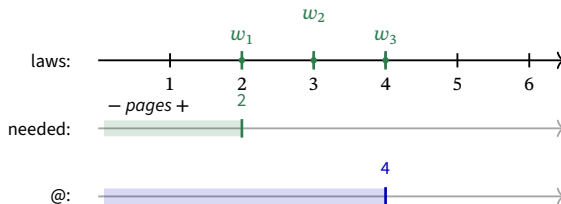
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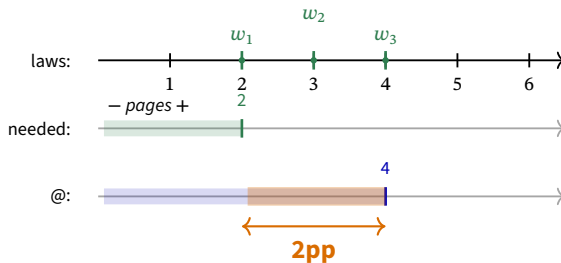
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Differentials + disjunction in *than*-clauses

(45) Ann is at most 20 cm taller than either Beth or Cleo.

Differentials + disjunction in *than*-clauses

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$\rightsquigarrow$ *Ann is at most 20 cm taller than Beth & Ann is at most 20 cm taller than Cleo*

Differentials + disjunction in *than*-clauses

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- What the sketched analysis predicts:

Differentials + disjunction in *than*-clauses

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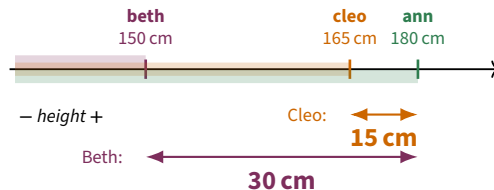


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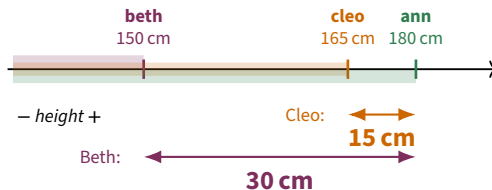


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- What the sketched analysis predicts:



- We predict the sentence is **true**, since it only compares Ann with the **taller** disjunct.

(Schwarzschild and Wilkinson 2002)

Part 8

Problems with quantifiers in *than*-clauses

Every in *than*-clauses

(46) Ann is taller than every boy is.

Every in than-clauses

- (46) Ann is taller than every boy is.
 $\rightsquigarrow$ for every boy x , Ann is taller than x

Every in *than*-clauses

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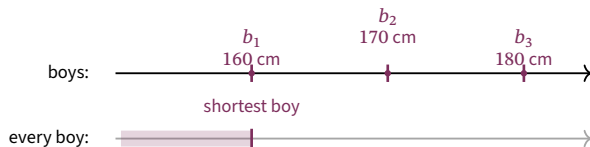


Every in *than*-clauses

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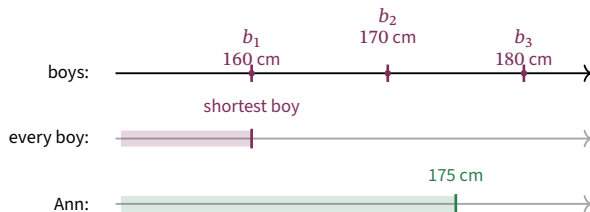


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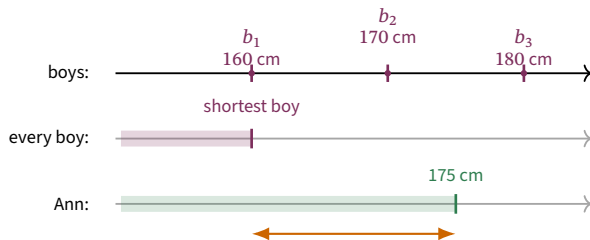


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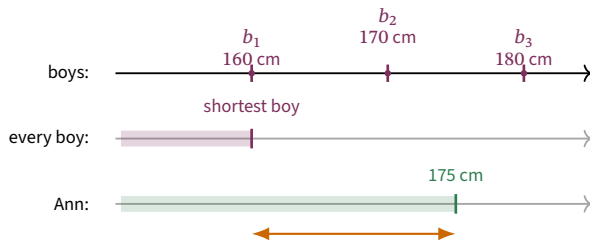
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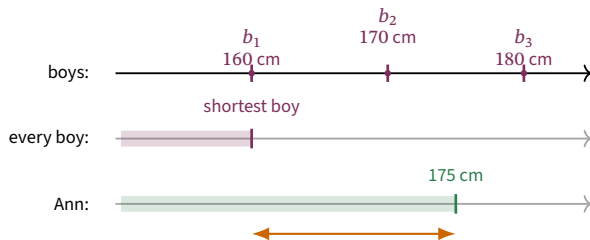


$\Rightarrow$ The analysis compares Ann only with the **shortest boy**.

Every in *than*-clauses

- (46) Ann is taller than every boy is.
 $\rightsquigarrow$ for every boy x , Ann is taller than x

- What the sketched analysis predicts:



$\Rightarrow$ The analysis compares Ann only with the **shortest boy**.

- Perhaps *every* can scope outside of the *than*-clause?

Expect in *than*-clauses

(47) Ann is taller than I predicted.

Expect in than-clauses

(47) Ann is taller than I predicted.

- **Context:** I predicted that Ann's height would be between 180 and 185 cm.

Expect in *than*-clauses

(47) Ann is taller than I predicted.

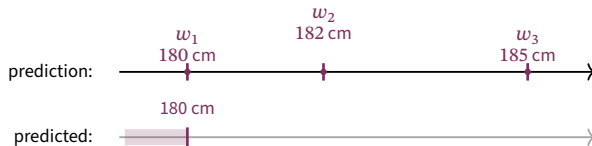
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Expect in *than*-clauses

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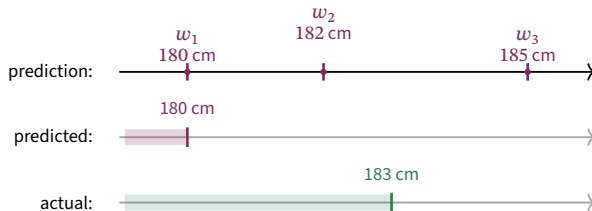
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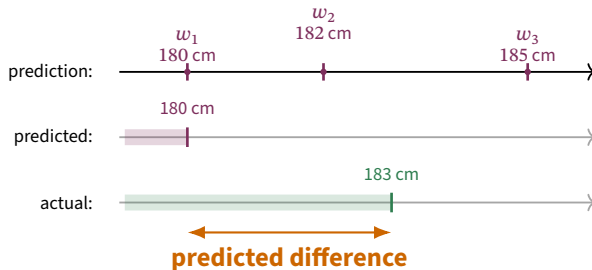
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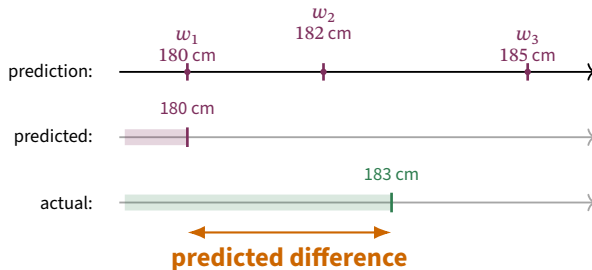
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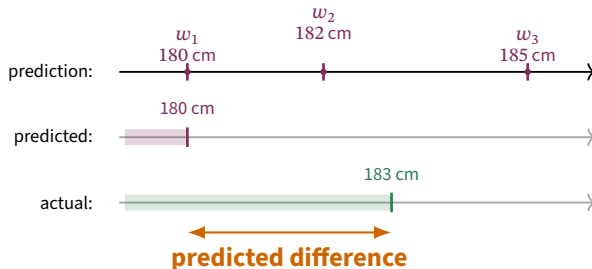


⇒ The analysis compares Ann's actual height with the **minimal** prediction.

Expect in *than*-clauses

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- **Context:** I predicted that Ann's height would be between 180 and 185 cm.
- What the sketched analysis predicts:



⇒ The analysis compares Ann's actual height with the **minimal** prediction.

- Here it wouldn't make sense to say *predict* QRs out of the *than*-clause...

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